NAVID BIN HASAN

412-606-8878 ◇ navidhasan0@gmail.com ◇ Pittsburgh, Pennsylvania

Website ◇ GitHub ◇ Google Scholar

Research Interests: Computational Biology, Bioinformatics

EDUCATION

Carnegie Mellon University Aug 2025 - Present
Ph.D. in Computational Biology (CPCB Program)

Bangladesh University of Engineering and Technology February 2025
M.Sc. in Computer Science and Engineering

- CGPA: 4.0/4.0
- Thesis: Leveraging and Correcting Weighted Quartet Distributions for Enhanced Species Tree Inference from Genome-Wide Data

Bangladesh University of Engineering and Technology May 2022
B.Sc. in Computer Science and Engineering

- CGPA: 4.0/4.0
- University Gold Medal Awardee (awarded to the highest CGPA holder of the batch)
- Prime Minister Gold Medal Awardee (awarded to the highest CGPA holder of each faculty of the batch)
- University Merit Scholarship and Dean's List Award in Each Year

RESEARCH EXPERIENCE

- **Graduate Research Assistant, CPCB Program, Carnegie Mellon University (August 2025 - Present)**
  Working under Dr. Yun William Yu on:
  - Developing novel algorithms to solve problems in computational biology and bioinformatics
  - Assessing the vulnerabilities in "Secure" Read Mapping algorithms
- **Leveraging and Correcting Weighted Quartet Distributions for Enhanced Species Tree Inference from Genome-Wide Data (April 2022 - Present)**
  Working under Dr. Md. Shamsuzzoha Bayzid on:
  - Introducing and exploring a wide array of methods for inferring weighted quartets and critically assessing their impact on the downstream species tree estimation
  - Developing a novel heuristic algorithm to identify patterns of Gene Tree Estimation Error in a weighted quartet distribution and adjust/correct the weights of the quartets accordingly to account for error
- **Branch Support Estimation on Distance-Based Phylogenetic Placements (March 2021 - April 2022)**
  Working under Dr. Siavash Mirarab and Dr. Md. Shamsuzzoha Bayzid on:
  - Implementing methods to calculate branch support for distance-based phylogenetic placements using non-parametric bootstrapping; integrating it to the popular phylogenetic placement tool APPLES [GitHub]
  - Devising and implementing a fast linear-algebraic implementation
- **Enhanced Partitioning for the QFM Method (October 2024 - Present)**
  Working under Dr. Mohammad Saifur Rahman on:
  - Exploration of alternate heuristic scoring functions to enhance the partitioning algorithm of QFM
- **Automatic Test Case Generation and Execution for Android Applications using Machine Learning (July 2023 - January 2024)**
  Working as an RA under Dr. Anindya Iqbal and in collaboration with Samsung Research Bangladesh on:
  - Automated black box testing of Android apps using UI exploration

- Automatic test case execution and verdict generation for Android applications using LLMs

- **Automatic High-Level Test Case Generation from Use Case with Large Language Models (January 2024 - November 2024)**
  Working under Dr. Anindya Iqbal on:
  - Constructing a novel use case to test case dataset
  - Fine-tuning LLMs with the dataset to generate accurate high-level test cases from use cases

PUBLICATIONS

- **Navid Bin Hasan**, Md. Ashraful Islam, Junaed Younus Khan, Sanjida Senjik, Anindya Iqbal, "Automatic High-Level Test Case Generation using Large Language Models," *MSR, 2025*
- **Navid Bin Hasan** et al., "Leveraging weighted quartet distributions for enhanced species tree inference from genome-wide data," *Genome Biology and Evolution*, 2025
- **Navid Bin Hasan**, Sohaib, Md. Shamsuzzoha Bayzid, "QT-WEAVER: Correcting quartet distribution improves phylogenomic analyses despite gene tree estimation error," *RECOMB-CG, 2025*
- **Navid Bin Hasan**, Metin Balaban, Avijit Biswas, Md. Shamsuzzoha Bayzid, Siavash Mirarab, "Distance-Based Phylogenetic Placement with Statistical Support," *Biology*, 11(8):1212, 2022
- **Navid Bin Hasan**, Avijit Biswas, Metin Balaban, Siavash Mirarab, Md. Shamsuzzoha Bayzid, "Fast and accurate branch support calculation for distance-based phylogenetic placements," *RECOMB-CG, 2022*

MANUSCRIPTS UNDER REVIEW

- **Navid Bin Hasan**, Sohaib, Md. Shamsuzzoha Bayzid, "Correcting quartet distribution improves phylogenomic analyses despite gene tree estimation error," under review in *Systematic Biology*

WORK EXPERIENCE

CPCB Program, Carnegie Mellon University — Pittsburgh, PA
Graduate Research Assistant — August 2025 - Present

Department of CSE, Bangladesh University of Engineering and Technology — Dhaka
Assistant Professor (on study leave to pursue PhD) — June 2025 - July 2025

Department of CSE, Bangladesh University of Engineering and Technology — Dhaka
Lecturer — July 2022 - June 2025

Department of CSE, Brac University — Dhaka
Lecturer — May 2022 - July 2022

Samsung Research Bangladesh-BUET Collaboration Research Project — Dhaka
Part-time Research Assistant — July 2023 - January 2024

ADDITIONAL EXPERIENCE

- Reviewer at International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW), 2024
- Juror at Blockchain Olympiad Bangladesh 2024 (Artificial Intelligence Category)
- Technical team member on the Class XI admission system under the IEIMS project conducted by BANBEIS, Bangladesh, 2024
- Technical team member on Conducting question set generation, result processing, allocation, and report generation for the Medical and Dental Admission Tests, Bangladesh, 2024
- Member and coordinator, Programming Contest Committee, Department of CSE, BUET
- Team member in investigating a critical data loss incident in a government construction permit system, 2023